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International Mathematics Assessments for Schools

2020 ~ 2021 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2019 ~ 2020 MIDDLE PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. Which of the following numbers has the largest value?
（A）20201　（B）20210　（C）22100　（D）22010　（E）22001

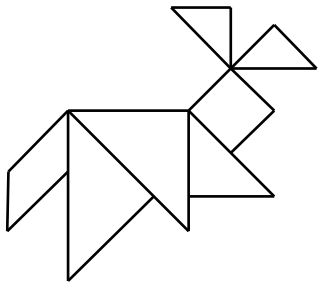
2. If Mike has $45 and Scott has $54, how much more money does Scott have compared to Mike?
（A）1　（B）4　（C）5　（D）9　（E）11

3. What is the numerical value of $20-19+2\times0\times2\times0-20+21$?
（A）0　（B）1　（C）2　（D）5　（E）6

4. Which of the following numbers is a multiple of 3?
（A）134　（B）178　（C）251　（D）577　（E）672

5. Harry arrived at school at 07:26 in the morning, which is 19 minutes before the start of the first class. What time does the first class start in the morning?
（A）07：35　（B）07：45　（C）07：50　（D）07：55　（E）08：05

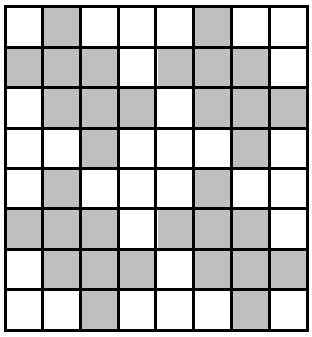
6. The diagram below shows a pattern that is made up using tangram puzzle pieces. How many more triangles than quadrilaterals are there in the diagram?



（A）2　（B）3　（C）4　（D）5　（E）7

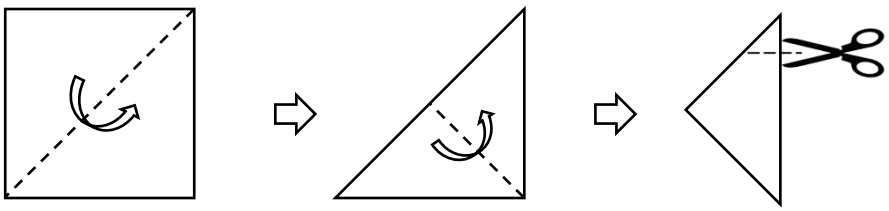
7. Jack miscalculated 25×48 as $25\times4\times8$. Which of the following operations must he select in order to arrive at the correct answer?
（A）Divide by 4　（B）Divide by 8　（C）Multiply by 2
（D）Add 100　（E）Multiply by $\frac{3}{2}$

8. In the figure below, the area of each small square is 1 cm^2. What is the total area, in cm^2, of the shaded region?



（A）24　（B）28　（C）32　（D）36　（E）40

9. Fold a square piece of paper diagonally twice and then cut a small corner as shown in the diagram below. After performing the task, completely unfold the paper. How many sides does our newly obtained figure have?



（A）6　（B）8　（C）10　（D）12　（E）16

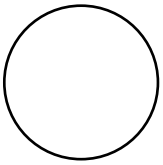
10. What is the units digit of $1+11+111+...+\underbrace{111...111}_{2020\ 1\text{'s}}$?

（A）0　（B）1　（C）2　（D）5　（E）6

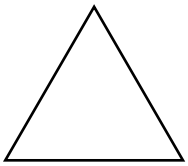
Questions 11-20, 4 marks each

11. Among the figures below, which has the most number of lines of symmetry?

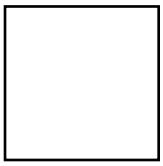
（A）



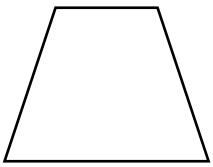
（B）



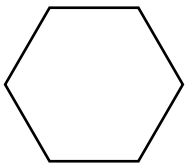
（C）



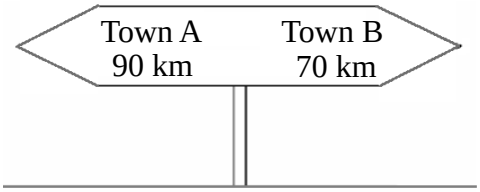
（D）



（E）



12. Albert drove from Town A to Town B at a constant speed of 80 km per hour. If the street sign shown below was seen during the trip, how many hours does it take Albert to drive from Town A to Town B?

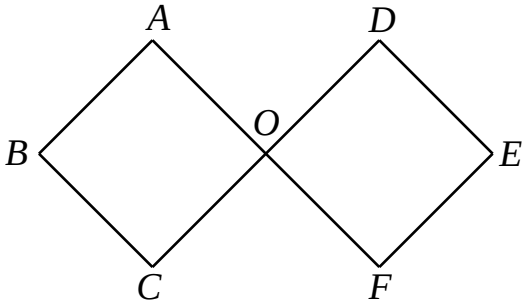


（A）2　（B）2.5　（C）3　（D）1.125　（E）0.875

13. Among the shapes listed below, which cannot always be formed by using two identical triangles?

（A）Rhombus　（B）Trapezoid　（C）Square
（D）Rectangle　（E）Parallelogram

14. In the figure below, it is known that the side length of both squares is 2 cm. An ant, whose initial position is at point *O* starts crawling continuously for 2020 cm along the following path: "*O-A-B-C-O-D-E-F-O-A-B-C-O-D-E-F-O-A*…". After completing the task, where will the ant arrive?



（A）Point *B*　（B）Point *E*　（C）Point *D*　（D）Point *C*　（E）Point *F*

15. The diagram below shows a clock displaying the time 10：10, where the hour hand points between the numbers 10 and 11, the minute hand points exactly to the number 2 and the second hand points to exactly to number 12. After exactly 50 minutes, what is the sum of the numbers pointed by the hour, minute, and second hands?

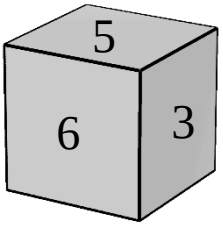


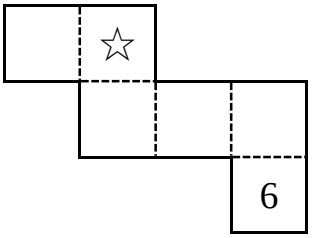
（A）24　（B）28　（C）29
（D）35　（E）None of the above

16. Given that the sum of the three smallest numbers among 11 consecutive positive integers is 63, what is the largest number among these 11 numbers?

（A）11　（B）21　（C）22　（D）30　（E）31

17. The figure on the lower left shows a cube, where the numbers from 1 to 6 are written on each face such that the sum of the numbers in any two opposite faces are all equal. If the figure on the lower right shows the diagram of the flattened view of the same cube, what is the number that is represented by ☆?





（A）1　（B）2　（C）3　（D）4　（E）5

18. In the expression below, letters *A*, *B*, *C*, *D* and *E* represent five different digits from 0 to 9. Which of the following letters has the largest value?

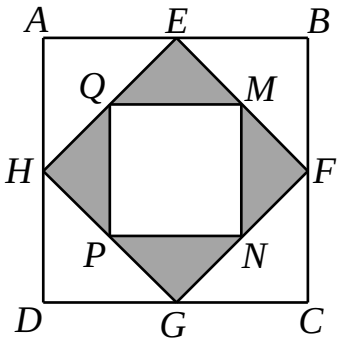
$$\begin{array}{cccccc} & A & 8 & B & 4 & C \\ - & 4 & D & 6 & E & 2 \\ \hline & 1 & 3 & 5 & 7 & 9 \end{array}$$

（A）*A*　（B）*B*　（C）*C*　（D）*D*　（E）*E*

19. There are 23 identical beads in a drawer, of which 8 are white and 15 are black. If you blindly take out the beads one at a time from the drawer, at least how many must be taken out to ensure that there are more black beads than white beads?

（A）1　（B）8　（C）15　（D）17　（E）23

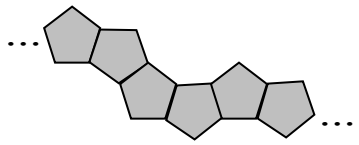
20. In the figure below, *ABCD* is a square with an area of 2020 cm^2. Connect the midpoints of each side to get square *EFGH* and then connect the midpoints of each side of square *EFGH* to get square *MNPQ*. What is the area, in cm^2, of the shaded region?



（A）400　（B）404　（C）450　（D）500　（E）505

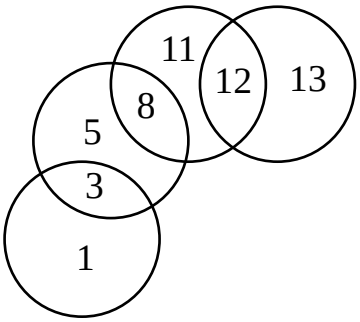
Questions 21-25, 6 marks each

21. In a park, there is a viewing trail that was paved with identical regular pentagonal stone bricks such that one side of a brick is connected to one side of another brick, as shown in the diagram. The gardener plans to make fences to each side of these stone bricks that are not in contact with the other stone bricks with a wooden board. If a total of 2021 wooden boards are needed to pave this trail, how many stone bricks are used?

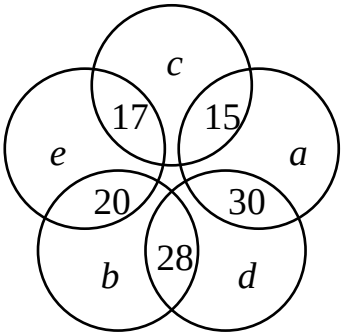


22. A school conducted a survey to all of its 216 students in the fifth grade. The results were as follows: 100 does not like to read comic books, 85 does not like to read novels and 33 does not like to read both comic books and novels. How many students like to read both comic books and novels?

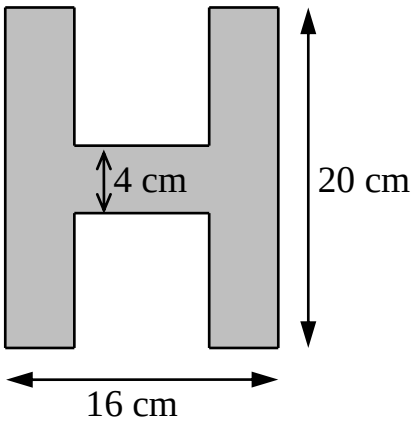
23. Some overlapping circles divide the plane into several regions. Numbers are filled in these areas such that for any pair of overlapping circles, say A and B, the number in the intersection area of circle A and B is the average of the two numbers in the areas solely in A or B. For example, the figure below shows $3=\frac{1+5}{2}$, $8=\frac{5+11}{2}$ and $12=\frac{11+13}{2}$.



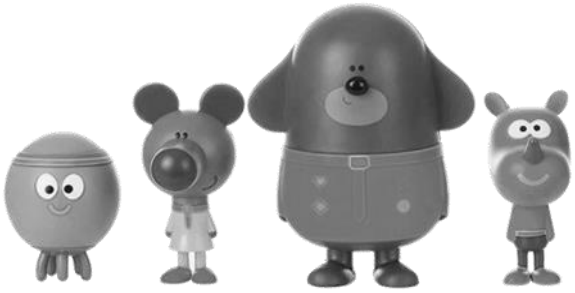
In the figure below, there are five overlapping circles and ten numbers are filled according to the conditions above. What is the sum of the numbers *a* and *b*?



24. The figure below shows one *H*-shaped paper, where each corner is a right angle. What is the perimeter, in cm, of the shaded region?



25. Grace wants to buy a total of 10 puppets, which have four kinds namely: Octopus, Mouse, Rhino and Dog puppet using the $700 that she has. The unit prices of these four puppets are $30, $50, $70 and $110, respectively. If she must buy at least one type of each puppet, how many different purchase combinations are there in total?



* * *